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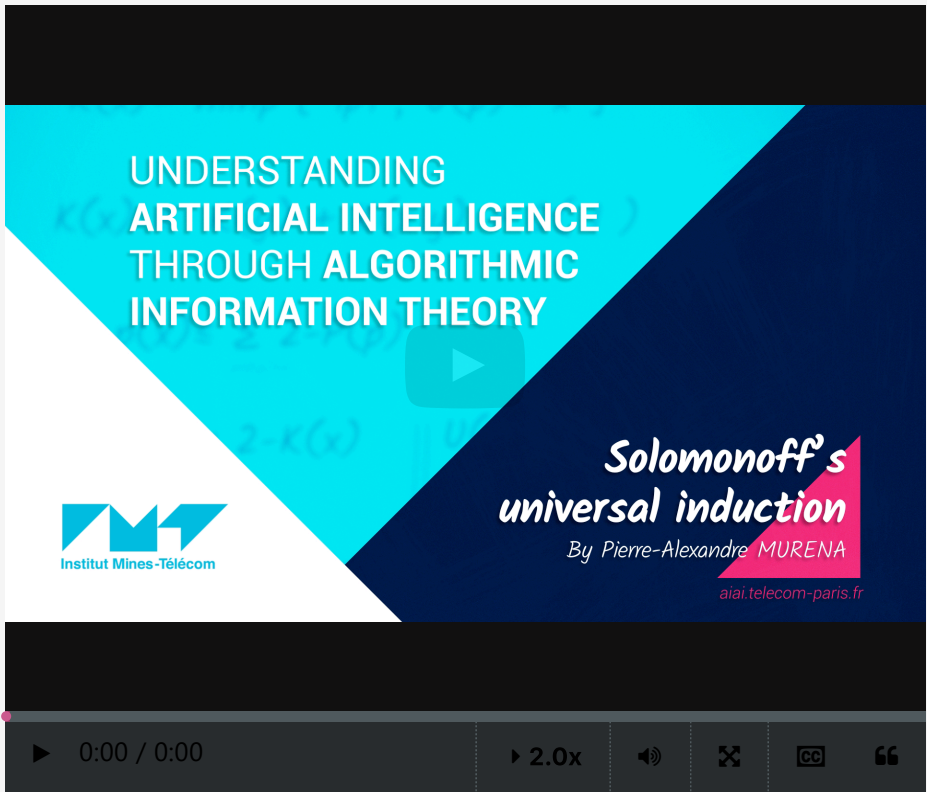
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Universal distribution

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Problem: Is there a link between Bayes Rule and Solomonoff's induction principle?

Solomonoff's universal induction



Start of transcript. Skip to the end.

Solomonoff's universal induction

Hi.

Today, I'll be talking about universal induction and...

spoiler alert...

there's a happy ending!

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hypothesis for a
what comes next.

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eterministic case, which
produced by a program p .

t the next symbol will be

s the sum of the
...

Prediction

0/1 point (ungraded)

What would be a good way to bet on x_n knowing the sequence x up to x_{n-1} , among the following?

- ☐ Choose the hypothesis that minimizes the complexity of x_n .
- ☐ Choose the hypothesis that minimizes the complexity of $x_{1:n}$.
- ☒ Choose the hypothesis that minimizes the complexity of $x_{1:n-1}$.

☐ Choose the hypothesis that minimizes the complexity of $x_{1:n+1}$.

☐ Choose the hypothesis that minimizes the complexity of $x_{1:\infty}$.

✖

Answer

Incorrect:

None of these choices is optimal. You may choose x_n that minimizes $K(x_n|x_{1:n-1})$ or, equivalently, $K(x_{1:n})|x_{1:n-1}$ (options b). Option e is closer to the optimum: minimize $K(x_{1:\inf})$ (with $K(x_{1:n-1})$ instantiated to known values) is an acceptable answer. However, this is only an approximation of the ideal choice, as one should consider the cumulative probability of all programs that generate $x_{1:\inf}$ (see Li and Vitányi 1993, Chapter 4.5.4).

Submit

You have used 1 of 1 attempt

Suggested reading:

- Li, M. & Vitányi, P. (1993). *An introduction to Kolmogorov complexity and its applications (3rd ed.)*. New York: Springer Verlag, ed. 1997, **Chapters 4.3 and 5.1**.

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